Leaf Disease Identification Using Image Processing and CNN

A Project Based Learning Report Submitted in partial fulfilment of the requirements for the award of the degree

of

Bachelor of Technology

in The Department of CSE

PRINCIPLES OF DEEP NEURAL COMPUTATION-24GMI3101HF

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JAN - 2026.

Introduction

Agriculture is one of the most important sectors that supports food production and the global economy. However, plant diseases are a major problem that reduce crop quality and yield. Early detection of leaf diseases is essential to prevent the spread of infection and minimize losses. Traditionally, farmers identify plant diseases through manual observation, which can be time-consuming and sometimes inaccurate, especially without expert knowledge.

With the advancement of technology, image processing and deep learning techniques provide an effective solution for automatic plant disease detection. Convolutional Neural Networks (CNNs) are widely used in image classification tasks because they can automatically learn important features from images. By training a CNN model on leaf images, it becomes possible to classify whether a leaf is healthy or affected by a specific disease. This system reduces the dependency on expert knowledge and provides a faster method for disease identification. By using digital images of plant leaves, the model can analyse patterns, colours, and textures that indicate different types of infections. The automation of this process saves time and ensures consistent results compared to manual inspection.

The proposed solution can also be implemented as a mobile or web-based application, making it easily accessible to farmers. With continuous improvements and larger datasets, the accuracy of the model can be further enhanced. Such intelligent systems contribute to smart agriculture and support sustainable farming practices.

In this project, leaf images are collected and processed using image processing techniques such as resizing and normalization. A CNN model is then trained to identify and classify different leaf diseases. This system can help farmers detect diseases at an early stage, reduce crop loss, and improve agricultural productivity through accurate and fast disease identification.

Literature Review/ Application Survey

Research paper-1: Using Deep Learning for Image-Based Plant Disease Detection

Authors: S. P. Mohanty, D. P. Hughes, M. Salathé

This research paper is one of the most influential studies in the field of plant disease detection using deep learning techniques. The authors focused on developing an automated system to identify plant diseases from leaf images using Convolutional Neural Networks (CNNs). For their experiments, they used the PlantVillage dataset, which contains more than 50,000 images of healthy and diseased plant leaves across multiple crop species.

In this study, several deep learning architectures such as AlexNet and GoogleNet were trained and tested to classify plant diseases. The models were evaluated under controlled conditions where leaf images had uniform backgrounds. The results showed very high classification accuracy, demonstrating that CNN models are capable of automatically learning important visual features such as color variations, texture patterns, and shape differences without manual feature extraction.

The research highlighted that deep learning significantly outperforms traditional machine learning methods, which require manual feature engineering. However, the authors also discussed certain limitations. Since the dataset images were captured under laboratory conditions, the trained models may not perform equally well in real-field environments where lighting, shadows, and complex backgrounds exist.

Overall, this paper laid the foundation for many future studies in plant disease detection and proved that deep learning-based approaches can provide accurate, fast, and automated solutions for agricultural applications.

Research Paper-2: Plant Leaf Disease Detection Using Transfer Learning

Authors: Various Researchers

This research paper focuses on improving plant disease classification using transfer learning techniques. Instead of training a Convolutional Neural Network (CNN) model from scratch, the researchers used pretrained deep learning models such as ResNet, VGG16, Inception, and MobileNet. These models were originally trained on large image datasets like ImageNet and later fine-tuned using plant leaf disease datasets.

The main objective of this study was to reduce training time and improve classification accuracy, especially when the available agricultural dataset is limited. Transfer learning allows the model to reuse previously learned image features such as edges, textures, and shapes, which are useful for identifying disease patterns on leaves. The researchers applied preprocessing techniques including image resizing, normalization, and data augmentation to improve model performance.

The experimental results showed that transfer learning models achieved very high accuracy compared to traditional machine learning approaches. In many cases, pretrained CNN models outperformed models trained from scratch. Additionally, MobileNet and other lightweight architectures demonstrated good performance with lower computational requirements, making them suitable for mobile and real-time applications.

The study also highlighted that transfer learning is highly effective for practical agricultural systems, as it requires less training data and computational power. However, performance may decrease when applied to real-world field images with complex backgrounds and lighting variations.

Overall, this research proves that transfer learning is a powerful and efficient technique for plant leaf disease detection and plays a significant role in developing smart agriculture solutions.

Research Paper-3: Real-Time Plant Disease Detection Using YOLO

Authors: Recent Deep Learning Researchers

This research paper focuses on real-time plant disease detection using object detection algorithms, particularly the YOLO (You Only Look Once) model. Unlike traditional classification models that only predict the disease type, this approach detects and localizes the infected regions within the leaf image. The main objective of this study was to develop a system that can identify plant diseases in real agricultural field conditions.

In this research, leaf images were collected under natural lighting with complex backgrounds. The YOLO model was trained to detect diseased spots and draw bounding boxes around the affected areas. Data preprocessing techniques such as image resizing, annotation of infected regions, and augmentation were performed to improve detection accuracy. The model was evaluated using metrics such as precision, recall, and mean Average Precision (mAP).

The results demonstrated that YOLO-based models provide fast and accurate detection, making them suitable for real-time applications. Compared to traditional CNN classification models, object detection models offer better practical usability because they show exactly where the disease is present on the leaf. This makes the system more useful for farmers and agricultural experts.

However, the study also mentioned challenges such as the need for properly annotated datasets and high computational resources during training. Despite these challenges, the research concludes that object detection techniques like YOLO are highly effective for smart farming systems, drone-based monitoring, and automated crop health assessment.

Research Paper-4: Comparative Study of Machine Learning and Deep Learning Approaches for Plant Disease Detection

Authors: Multiple Researchers

This research paper presents a comparative analysis between traditional machine learning techniques and modern deep learning approaches for plant leaf disease detection. The primary objective of the study was to evaluate the performance of both methods and determine which approach provides better accuracy and reliability under different conditions.

In the traditional approach, image processing techniques were first applied for preprocessing tasks such as noise removal, segmentation, and color space conversion. After preprocessing, features like color, texture, and shape were extracted using techniques such as Gray-Level Co-occurrence Matrix (GLCM), Histogram of Oriented Gradients (HOG), and color histograms. These features were then classified using machine learning algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forest.

For the deep learning approach, Convolutional Neural Networks (CNNs) were used to automatically learn features directly from raw leaf images. The study found that CNN-based models significantly outperformed traditional methods in terms of accuracy and robustness. Deep learning models required less manual intervention and were better at handling variations in lighting, background, and leaf orientation.

The results concluded that although traditional machine learning methods are computationally less expensive, deep learning techniques provide higher accuracy and better generalization. Therefore, CNN-based systems are more suitable for practical plant disease detection applications in modern agriculture.

Research Paper-5: Lightweight CNN Models for Mobile-Based Plant Disease Detection

Authors: Various Researchers

This research paper focuses on developing lightweight Convolutional Neural Network (CNN) models for plant disease detection that can be deployed on mobile devices. The main objective of the study was to create a system that provides accurate disease identification while reducing computational complexity and memory usage. This is important because many farmers may rely on smartphones rather than high-performance computers.

In this study, lightweight deep learning architectures such as MobileNet, EfficientNet, and ShuffleNet were used. These models are designed to reduce the number of parameters and computational cost while maintaining high classification accuracy. The researchers trained the models using plant leaf image datasets and applied preprocessing techniques such as resizing, normalization, and data augmentation to improve model performance.

The experimental results showed that lightweight CNN models achieved high accuracy comparable to larger deep learning models. At the same time, they required less processing power and memory, making them suitable for real-time mobile applications. The system was able to classify healthy and diseased leaves quickly, providing instant results to users.

The study concluded that lightweight CNN models are highly practical for real-world agricultural applications. By integrating the model into a mobile application, farmers can capture leaf images and receive immediate disease diagnosis. This approach supports smart farming practices, reduces dependency on experts, and enables early disease detection, ultimately improving crop productivity and sustainability.

References

1. Mohanty, S.P., Hughes, D.P., & Salathé, M. (2016). *Using Deep Learning for Image-Based Plant Disease Detection*. Frontiers in Plant Science Classic, first large-scale PlantVillage work (introduces PlantVillage dataset & CNN approach).
   https://www.frontiersin.org/articles/10.3389/fpls.2016.01419/full

2. Hassan, S.M. et al. (2022). *A Survey on Different Plant Diseases Detection Using Machine/Deep Learning* (MDPI Electronics) — compares handcrafted vs deep-learning pipelines.
   https://www.mdpi.com/2079-9292/10/12/1388

3. Falaschetti, L. et al. (2022). *A CNN-based image detector for plant leaf diseases* — describes resource-constrained CNNs and embedded deployment. Useful for real-time / low-power discussion.
   https://www.sciencedirect.com/science/article/pii/S2468067222001080

4. Transfer-learning / ResNet50 application (case study). *Transfer Learning Based Plant Diseases Detection Using ResNet50* — good example of applying pretrained CNNs (ResNet) to plant disease datasets.
   https://www.researchgate.net/publication/340691402_Transfer_Learning_Based_Plant_Diseases_Detection_Using_ResNet50

5. PlantVillage dataset (official TFDS / Kaggle copies) — primary benchmark dataset used across many papers (≈54k images; 38 classes). Cite this when discussing datasets.
   https://www.tensorflow.org/datasets/catalog/plant_village